

Practical:4

Implementation and Time analysis of factorial program using iterative and recursive method

/08/2026

Aim: Implementation and Time analysis of factorial program using iterative and recursive method

Algorithm:

Algorithm

Input:- Enter a number
Output:- Display factorial of element

Step1:- Start

Step2:- Read the value of n

Step3:- Iterative method

```

fact = 1;
for (int i = 1; i <= n; i++)
    Then fact = fact * i;
return fact
    
```

Step4:- Recursive method

```

Read the value of n
if n == 0 || n == 1
    Then return 1;
return n * factorial (n-1);
    
```

Step5:- if (n < 0) factorial is not defined for negative number

Step6:- Display both iterative & Recursive factorial of n

Step7:- Stop

Program:

```
#include <iostream>
```

```
using namespace std;
```

```
int factorialIterative(int n)
```

```
{
```

```
    int fact = 1;
```

```
    for (int i = 1; i <= n; i++)
```

```
    {
```

```
        fact = fact * i;
```

```
    }
```

```
    return fact;
```

```
}
```

```
int factorialRecursive(int n)
```

```
{
```

```
    if (n == 0 || n == 1)
```

```
        return 1;
```

```
    return n * factorialRecursive(n - 1);
```

```
}
```

```
int main()
{
    int n;

    cout << "Enter a number: ";
    cin >> n;

    if (n < 0)
    {
        cout << "Factorial is not defined for negative numbers.";
        return 0;
    }
    int iterativeResult = factorialIterative(n);
    int recursiveResult = factorialRecursive(n);

    cout << "\nFactorial using Iterative Method = "<< iterativeResult ;

    cout << "\nFactorial using Recursive Method = "<< recursiveResult;

    return 0;
}
```

Output:

```

C:\Users\Admin\OneDrive\Pic x + v
Enter a number: 6

Factorial using Iterative Method = 720
Factorial using Recursive Method = 720
-----
Process exited after 14.29 seconds with return value 0
Press any key to continue . . . |

```

Time Complexity:

Time Complexity:-

```

int factorial (n) {
    if (n==0 || n==1) {
        return 1;
    }
    return n * factorial (n-1);
}

```

Worst case:-

$T(n) = n + (n-1) + (n-2) + \dots + 1$
 $T(n) = \frac{n(n+1)}{2}$
 $T(n) = O(n^2)$

for a large value of n the function calls itself n time before reaching the base case.

Average case:-

for normal value of n , approximately n recursive call are made.

$T(n) = O(n)$

Best case:-

$T(n) = O(1)$

$n=0$ (or) $n=1$ so the base is reached immediately.

Conclusion:

Both iterative and recursive methods correctly calculate factorial. Both take $O(n)$ time, but the iterative method uses less memory ($O(1)$) compared to recursion ($O(n)$). Thus, iteration is more memory-efficient, while recursion is easier to understand conceptually.